SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I Durgi Lokesh (192372216) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Durgi Lokesh*

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

(a) Representation Using Propositional Logic

A rule-based medical diagnosis system uses propositional logic to represent symptoms and diseases.

Symbols Used

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Fl	Patient has Flu
M	Patient has Measles
P	Patient has Pneumonia

Knowledge Base (KB)

Rule 1: If a patient has Fever and Cough, then the patient may have Flu.

Rule 2: If a patient has Fever and Rash, then the patient may have Measles.

Rule 3: If a patient has Cough and Breathlessness, then the patient may have Pneumonia.

Facts about Patient A

- Fever = True (**F**)
- Cough = True (**C**)
- Breathlessness = True (**B**)

Thus, the initial knowledge base is:

(b) Forward Chaining

Forward Chaining is a **data-driven inference technique** that starts with the known facts and repeatedly applies rules to derive new facts until no more rules can be applied.

Step 1: Initial Facts

Working Memory = **{F, C, B}**

Step 2: Apply Rule 1

Rule:

Since **F = True** and **C = True**, Rule 1 is satisfied.

Derived Fact: Fl (Possible Flu)

Updated Working Memory:

Step 3: Apply Rule 2

Rule:

Although **F = True**, the fact **Rash (R)** is not available.

Therefore, Rule 2 **cannot be applied**, and Measles is not inferred.

Step 4: Apply Rule 3

Rule:

Since **C = True** and **B = True**, Rule 3 is satisfied.

Derived Fact: P (Possible Pneumonia)

Updated Working Memory:

No additional rules can be applied, so the inference process stops.

Forward Chaining Summary

Step	Rule Applied	Result
1	Initial Facts	F, C, B
2	F ∧ C → Fl	Flu derived
3	F ∧ R → M	Cannot fire (Rash absent)
4	C ∧ B → P	Pneumonia derived

Final Diagnoses

- ✓ Possible Flu
- ✓ Possible Pneumonia
- ✗ Possible Measles (not inferred)

(c) Backward Chaining

Backward Chaining is a **goal-driven inference technique**. It starts with the goal and checks whether the available facts can prove it.

Goal

Prove:

The rule that concludes **P** is:

The system now checks the required sub-goals.

- Is **Cough (C)** true? → **Yes**
- Is **Breathlessness (B)** true? → **Yes**

Since both conditions are satisfied, the rule is successfully applied.

Therefore,

Hence, the goal **"Patient A has Pneumonia"** is proved.

Goal Reduction Tree

```
          Goal
       Pneumonia (P)
            |
       C ∧ B → P
        /     \
       /       \
  Cough (C)   Breathlessness (B)
```

```
 |            |
True        True
 \          /
  \________/
      |
 Goal Satisfied
```

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x$: Vehicle(x) $\land$ EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]

II. $\forall x$: ClearPath(x) $\rightarrow$ GreenSignal(x) $\land$ $\neg$RedSignal(x) [Rule 2]

III. $\forall x$ $\forall y$: Vehicle(x) $\land$ Behind(x,y) $\land$ GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

(a) Apply Unification

The traffic management system uses **First Order Logic (FOL)** to manage emergency vehicles. Unification is the process of matching variables in a rule with known facts to derive new conclusions.

Knowledge Base

Rule 1:

Rule 2:

Rule 3:

Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

Unification Process

Match Rule 1 with the given facts.

Predicate 1

Vehicle(x)

Fact:

Vehicle (Ambulance)

Substitution (MGU):

Predicate 2

EmergencyType(x)

Using the same substitution:

EmergencyType (Ambulance)

This matches the fact.

Therefore,

Applying Rule 1 gives:

ClearPath(Ambulance)

(b) Resolution Proof for Proceed(CarA)

Step 1: Convert Rules into CNF

Rule 1

CNF:

Rule 2

CNF:

Rule 3

CNF:

Resolution Steps

From the facts:

- Vehicle(Ambulance)
- EmergencyType(Ambulance)

Resolve with Rule 1:

⇒ **ClearPath(Ambulance)**

Resolve with Rule 2:

⇒ **GreenSignal(Ambulance)**

Now use:

- Vehicle(CarA)
- Behind(CarA, Ambulance)
- GreenSignal(Ambulance)

Resolve with Rule 3:

⇒ **Proceed(CarA)**

Hence, the statement **Proceed(CarA)** is successfully proved.

Resolution Proof Tree

```
Vehicle(Ambulance)     EmergencyType(Ambulance)
         \                 /
          \               /
         ClearPath(Ambulance)
                 |
                 |
         GreenSignal(Ambulance)
                 |
     -------------------------------
    |            |             |
Vehicle(CarA) Behind(CarA,Ambulance)
    \            |            /
     \___________|___________/
                 |
            Proceed(CarA)
```

(c) Analysis

The current knowledge base works correctly for **one emergency vehicle**, but it is **not sufficient** when multiple emergency vehicles are present at the same time.

Additional Rules Required

1. **Priority Rule**

    - If two emergency vehicles are present, the one with higher priority (e.g., Ambulance over Fire Truck) should be given the green signal first.
2. **Conflict Resolution Rule**
    - If two emergency vehicles approach the same junction simultaneously, the system should avoid giving both directions a green signal.
3. **Traffic Coordination Rule**
    - After the first emergency vehicle passes, the next emergency vehicle should automatically receive a clear path.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

(a) Convert the Knowledge Base into Conjunctive Normal Form (CNF)

The agricultural expert system uses propositional logic to determine irrigation requirements and crop health.

Knowledge Base

P1: SoilDry → IrrigationNeeded

P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod

P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Facts

- SoilDry = True
- CropWheat = True

Symbols Used

Symbol	Meaning
S	SoilDry
I	IrrigationNeeded
W	CropWheat
D	ApplyDripMethod
R	CropAtRisk

CNF Conversion

P1:

Remove implication:

P2:

Remove implication:

Apply De Morgan's Law:

P3:

Remove implication:

Apply De Morgan's Law:

CNF Knowledge Base

1.
2.
3.
4.
5.

(b) Resolution Refutation to Prove ApplyDripMethod

Goal

Prove:

Negate the goal:

Add it to the knowledge base.

Resolution Steps

Step 1

From

-
-

Resolve to obtain:

Step 2

From

-
-

Resolve:

Step 3

From

-
-

Resolve:

Step 4

Resolve

-
-

Result:

(Empty Clause)

Since the empty clause is obtained, **ApplyDripMethod** is successfully proved.

Resolution Proof Tree

```
      SoilDry (S)
         |
     ¬S ∨ I
```

|

IrrigationNeeded (I)

|

¬I ∨ ¬W ∨ D

|

CropWheat (W)

|

ApplyDripMethod (D)

|

¬D

|

□

(c) Analysis if SoilDry is Removed

Suppose the fact **SoilDry (S)** is removed from the knowledge base.

Step 1

Without **S**, Rule P1 cannot derive **IrrigationNeeded (I)**.

Step 2

Since **I** is not available, Rule P2 cannot infer **ApplyDripMethod (D)**.

Step 3

Without proving **ApplyDripMethod**, there is no evidence that the irrigation method has been applied.

Step 4

Rule P3 requires **¬ApplyDripMethod** and **CropWheat** to conclude **CropAtRisk**. However, the knowledge base does **not** explicitly state **¬ApplyDripMethod**.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

(a) Construct the Agent's Belief State

The Wumpus World agent receives percepts from the environment and uses Propositional Logic with Modus Ponens to derive new knowledge.

Percepts

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Knowledge Base

1. Stench[1,2] → Wumpus is adjacent to [1,2]
2. Breeze[1,1] → Pit is adjacent to [1,1]
3. Glitter[x,y] → Gold at [x,y]
4. ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

Belief State

From the given percepts:

- Stench[1,2] = True
- Breeze[1,1] = True
- Glitter[2,2] = True

Applying Modus Ponens

Rule 1:

Stench[1,2] → Wumpus adjacent to [1,2]

Since Stench[1,2] is true,

Derived Fact: Wumpus is adjacent to [1,2].

Rule 2:

Breeze[1,1] → Pit adjacent to [1,1]

Since Breeze[1,1] is true,

Derived Fact: Pit is adjacent to [1,1].

Rule 3:

Glitter[2,2] → Gold at [2,2]

Since Glitter[2,2] is true,

Derived Fact: Gold is located at [2,2].

Derived Conclusions

- Wumpus is adjacent to [1,2]
- Pit is adjacent to [1,1]
- Gold is at [2,2]

(b) Forward Chaining

Forward Chaining starts with the known percepts and repeatedly applies rules.

Initial Facts

- Stench[1,2]
- Breeze[1,1]
- Glitter[2,2]

Inference Steps

Step 1

Apply Rule 1:

Stench[1,2]

↓

Wumpus adjacent to [1,2]

Step 2

Apply Rule 2:

Breeze[1,1]

↓

Pit adjacent to [1,1]

Step 3

Apply Rule 3:

Glitter[2,2]

↓
Gold at [2,2]
Possible Locations

- Possible Wumpus: Adjacent cells of [1,2], such as [1,1], [1,3], or [2,2].
- Possible Pit: Adjacent cells of [1,1], such as [1,2] or [2,1].
- Gold: Located at [2,2].

(c) Safety Analysis of Path
Path
Analysis

- At [1,1], the agent senses a Breeze, indicating that a Pit is nearby.
- At [1,2], the agent senses a Stench, indicating that the Wumpus is nearby.
- Although Glitter confirms that Gold is at [2,2], the path passes through cells that may contain dangerous hazards.
- The rule ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y] cannot be applied because the agent has not proved that these cells are free from both the Wumpus and Pit.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand